Kinematics in Two-Dimensions

Do everything we did in 1D kinematics, but with vectors!

Motion in 2D

- Particle path: “trajectory”
- Particle position: $\vec{r} = x\hat{i} + y\hat{j}$
- Particle displacement: $\Delta\vec{r} = \vec{r}_{\mathrm{f}} - \vec{r}_{\mathrm{i}}$
- Average velocity in 2D:

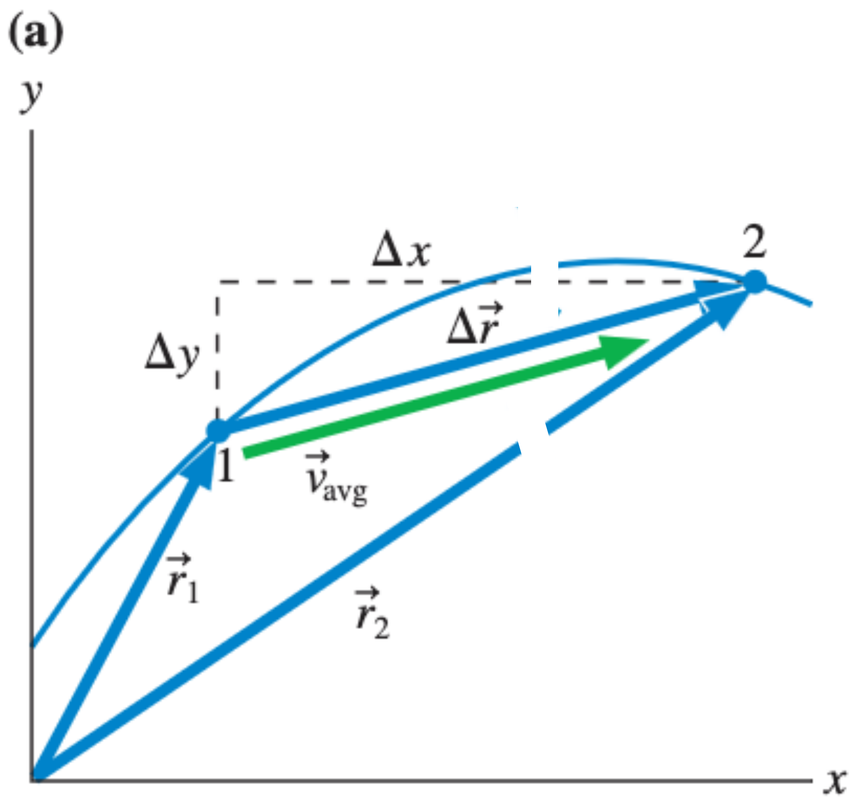
$$\vec{v}_{\mathrm{avg}} = \frac{\Delta\vec{r}}{\Delta t} = \frac{\Delta x}{\Delta t}\hat{i} + \frac{\Delta y}{\Delta t}\hat{j}$$

- Instantaneous velocity in 2D:

$$\vec{v}_{\mathrm{inst}} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}}\hat{i} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{j}$$

2D motion can be in any plane, but we’ll usually use x, y

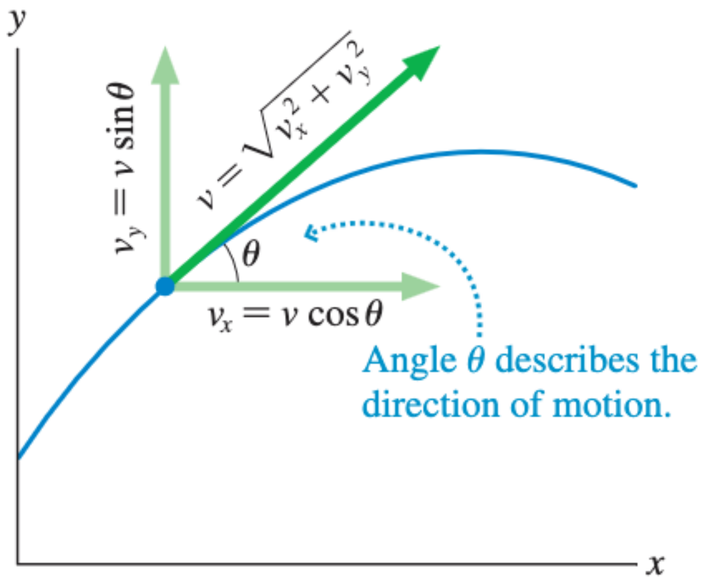
Velocity is a vector & tangent to trajectory



Velocity vector

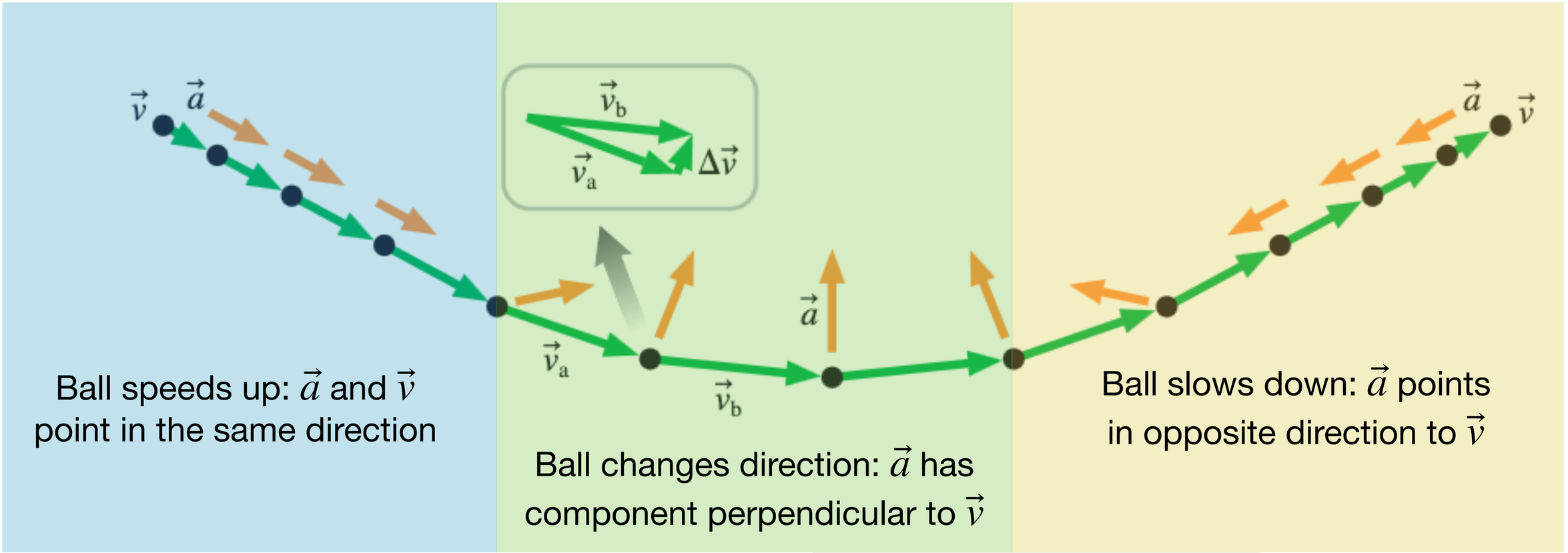
- Everything we learned about vectors applies to velocity vectors
  - $v_x = v\cos\theta$
  - $v_y = v\sin\theta$
  - $v = \sqrt{v_x^2 + v_y^2}$
  - $\theta = \arctan\left(\frac{v_y}{v_x}\right)$

- Particle speed at given point.
- Seed always positive.



Acceleration

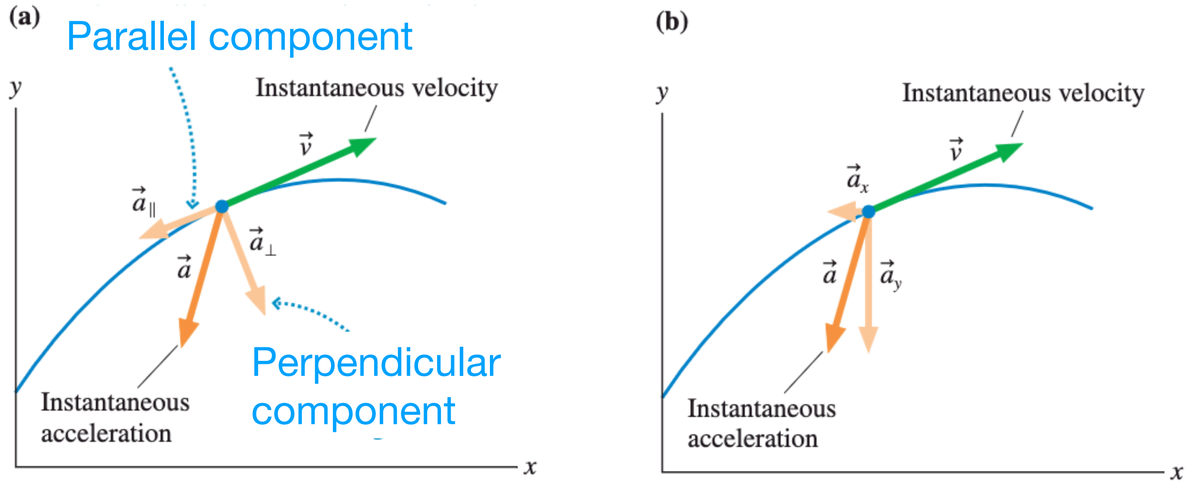
- Acceleration = change in velocity
- $a_{\text{avg}} = \dfrac{\Delta\vec{v}}{\Delta t}$ where $\Delta\vec{v} = \vec{v}_b - \vec{v}_a$
- $\vec{a}_{\text{inst}} = a_x\hat{i} + a_y\hat{j} = \dfrac{\mathrm{d}\vec{v}}{\mathrm{d}t} = \dfrac{\mathrm{d}v_x}{\mathrm{d}t}\hat{i} + \dfrac{\mathrm{d}v_y}{\mathrm{d}t}\hat{j}$
- **Recall:** velocity can change in two possible ways
  - Magnitude = change in speed (length of velocity vector changes)
  - Direction (orientation of velocity vector changes)
- **In 1D kinematics we only consider change in speed**


$\vec{v}$
$\vec{a}$
$\vec{v}_b$
$\vec{v}_a$
$\Delta\vec{v}$
$\vec{a}$
$\vec{v}_a$
$\vec{v}_b$
$\vec{a}$
$\vec{v}$
Ball speeds up: $\vec{a}$ and $\vec{v}$ point in the same direction
Ball changes direction: $\vec{a}$ has component perpendicular to $\vec{v}$
Ball slows down: $\vec{a}$ points in opposite direction to $\vec{v}$

Decomposing Acceleration

- Choices: (tangent, parallel), or (x, y)
- Tangent-/parallel-to-motion
  - Parallel component changes object's speed
  - Perpendicular component changes object's direction
  - good for circular motion
- Xy-decomposition mathematically convenient
  - Tangent/parallel can change direction
  - Good for projectile motion

Same acceleration vector, different decomposition



An object changing direction always has a component of acceleration perpendicular to the direction of motion

Constant Acceleration

- $\vec{a} = a_x \hat{i} + a_y \hat{j}$
  - components a_x and a_y are constant also
  - **Components of constant acceleration are independent** (but still connected through Δt)
- Everything 1D constant acceleration carries over to 2D
  - $x_\mathrm{f} = x_\mathrm{i} + v_{\mathrm{i}x}\Delta t + \frac{1}{2} a_x \Delta t^2$
  - $v_{\mathrm{f}x} = v_{\mathrm{i}x} + a_x \Delta t$
  - $y_\mathrm{f} = y_\mathrm{i} + v_{\mathrm{i}y}\Delta t + \frac{1}{2} a_y \Delta t^2$
  - $v_{\mathrm{f}y} = v_{\mathrm{i}y} + a_y \Delta t$



Projectile Motion

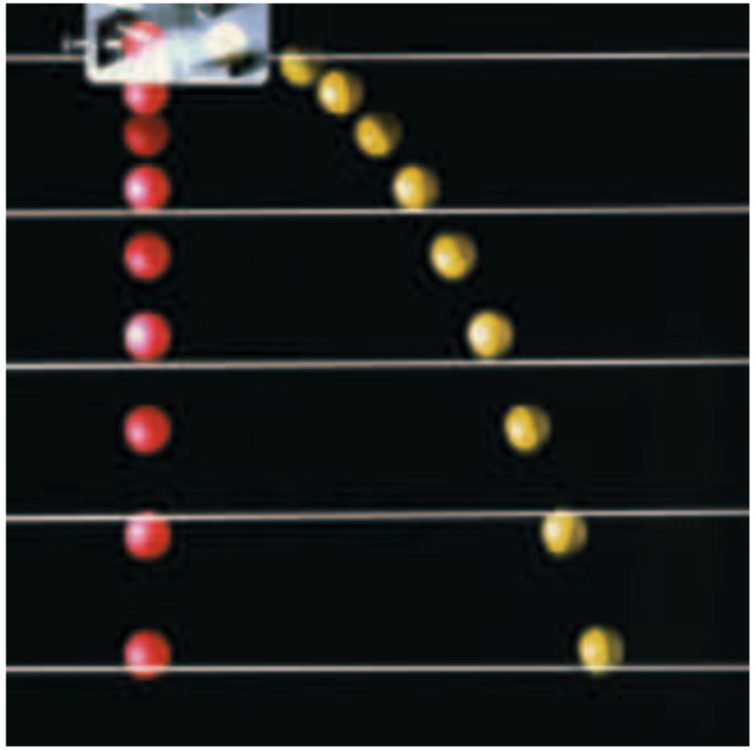
- Ball launched horizontally at height h. At same time, second ball dropped from same height. **Balls hit the ground simultaneously.**

- The horizontal and vertical components are independent

- Gravity acts downward, not sideways

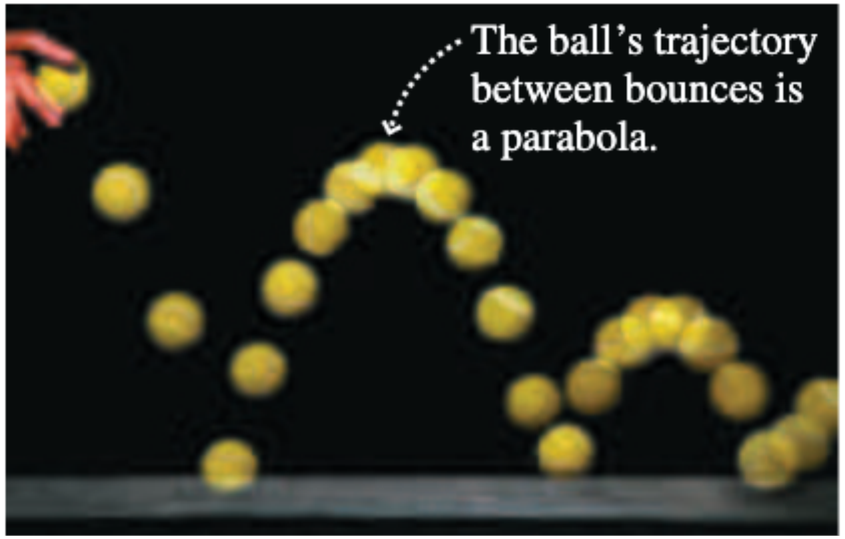
  - uniform motion at constant velocity (horizontal)

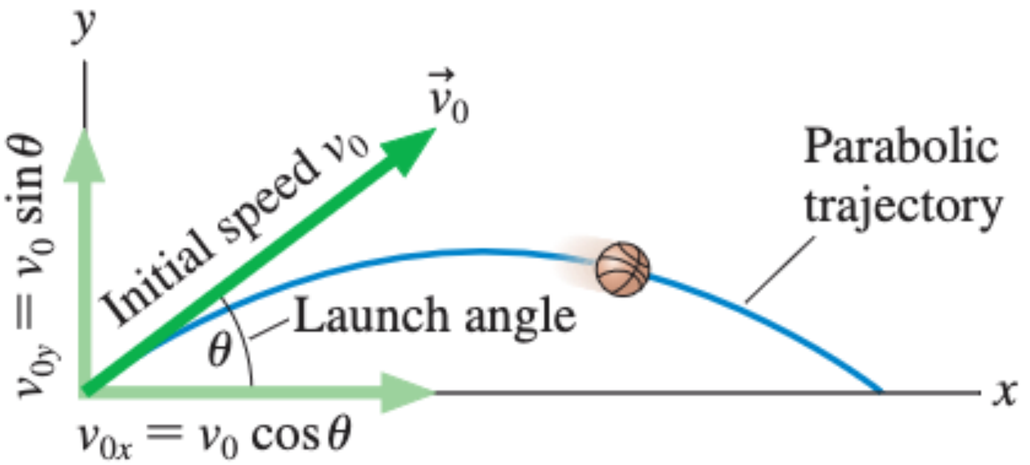
  - Free-fall motion (vertical)



Projectile Motion Model

- Extension of free-fall motion
- Projectile = **object that moves in 2D dimensions under the influence of gravity only**
- Neglect air resistance
- Initial velocity vector components:
  - $v_{0x} = v_0 \cos\theta$
  - $v_{0y} = v_0 \sin\theta$





Example

- A stunt man drives a car off a $10\,\mathrm{m}$ high cliff at a speed of $20\,\mathrm{m/s}$. How far does the car land from the base of the cliff?

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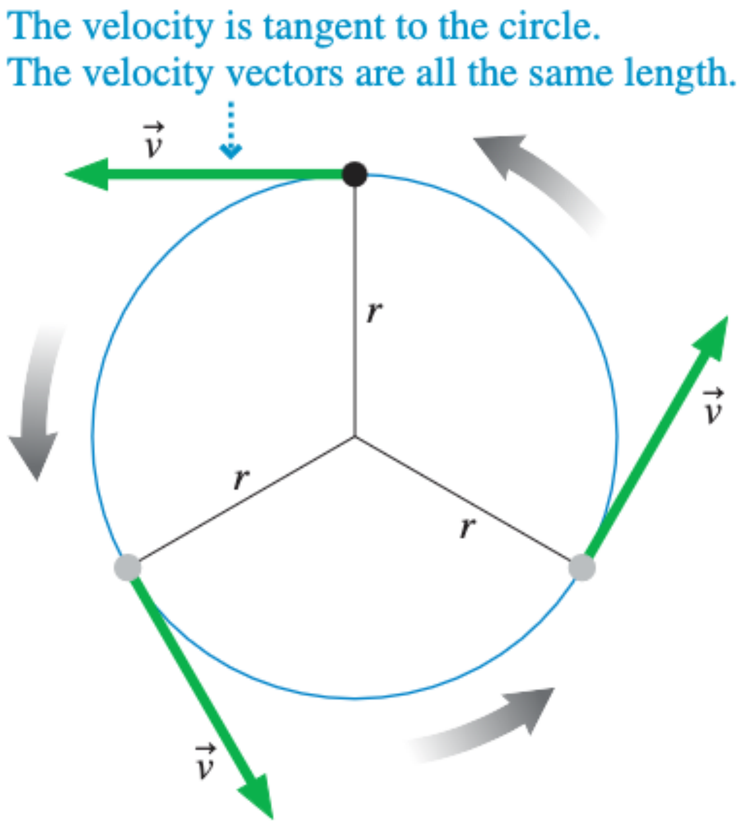
- $t_1 = \sqrt{\dfrac{2y_0}{g}} = \sqrt{\dfrac{2(10.0\,\mathrm{m})}{9.80\,\mathrm{m/s^2}}} = 1.43\,\mathrm{s}$

- $x_1 = v_0 t_1 = (20.0\,\mathrm{m/s})(1.43\,\mathrm{s}) = 28.6\,\mathrm{m}$

Circular Motion

Circular Motion Facts

- As long as $|\vec{r}|$ is constant, the velocity vector is always perpendicular to $\vec{r}$
  - Velocity is always purely tangential v_t
  - $v_t = r\omega$ (ω must be in $\mathrm{rad/s}$)
  - If $|\vec{r}|$ was not constant $\vec{v}$ would have a radial component.
- Acceleration for uniform circular motion is always perpendicular to $\vec{v}$
- Acceleration for nonuniform circular motion has a that is perpendicular to $\vec{v}$ and a part that is tangential to $\vec{v}$

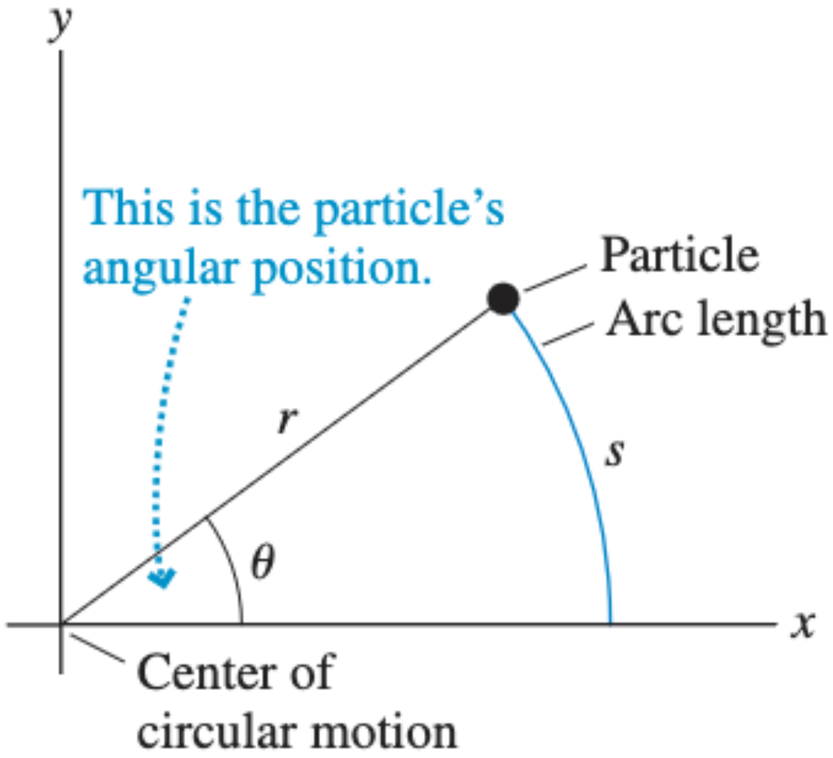


Terms

- Rev/revolution: one complete trip around the circle
- Period: time interval for particle to complete one rev; represented by T

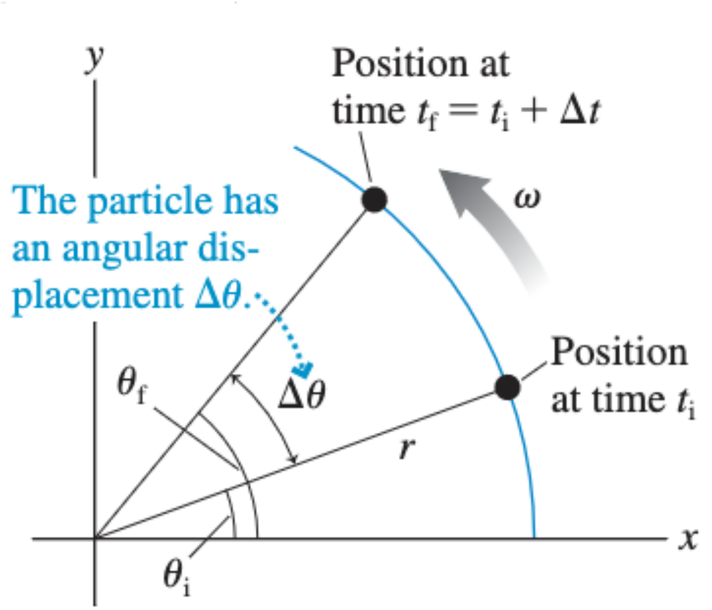
Angular Position

- More convent to describe position with magnitude r and direction θ
- **Angular position** is the angle θ
  - Usually measured the positive x-axis
  - Counterclockwise (ccw) positive
  - Clockwise (cw) negative
- Radians
  - $\theta_{\text{rad}} = \dfrac{s}{r}$
  - s (here) is the arc length (not a generic axis)
  - Arch length $s = r\theta_{\text{rad}}$
  - Full circle: $\theta_{\text{rad}} = 2\pi$



Angular Velocity

- Angular displacement: $\Delta\theta = \theta_f - \theta_i$
- Particle's circular motion = rate of change of θ
- Particle's linear motion = rate of change of arc length s
- Average angular velocity (rate of change of angular position)
  - $\omega = \dfrac{\Delta\theta}{\Delta t}$
    - ω is lowercase Greek omega (not ordinary w)
- Instantaneous angular velocity
  - $\omega = \lim\limits_{\Delta t \to 0} \dfrac{\Delta\theta}{\Delta t} = \dfrac{d\theta}{dt}$
- SI unit of angular velocity is $\dfrac{\text{rad}}{s}$
- Other common units are $\dfrac{\text{deg}}{s}, \dfrac{\text{rev}}{s}, \dfrac{\text{rev}}{\text{min}}, \text{rpm}$



Angular Velocity

- Angular velocity is 1D
- Can be positive or negative
- All of the linear (1D) motion results apply
  - ω is the slope of the θ vs. time graph

Uniform Circular Motion

- Another example of motion in a plane
- Particle that moves in a circle with constant speed
  - $v = \dfrac{1\ \text{rev}}{T} = \dfrac{2\pi r}{T}$
- Velocity vector $\vec{v}$ always tangent to the circle
- Speed v constant, so vector $\vec{v}$ always the same length

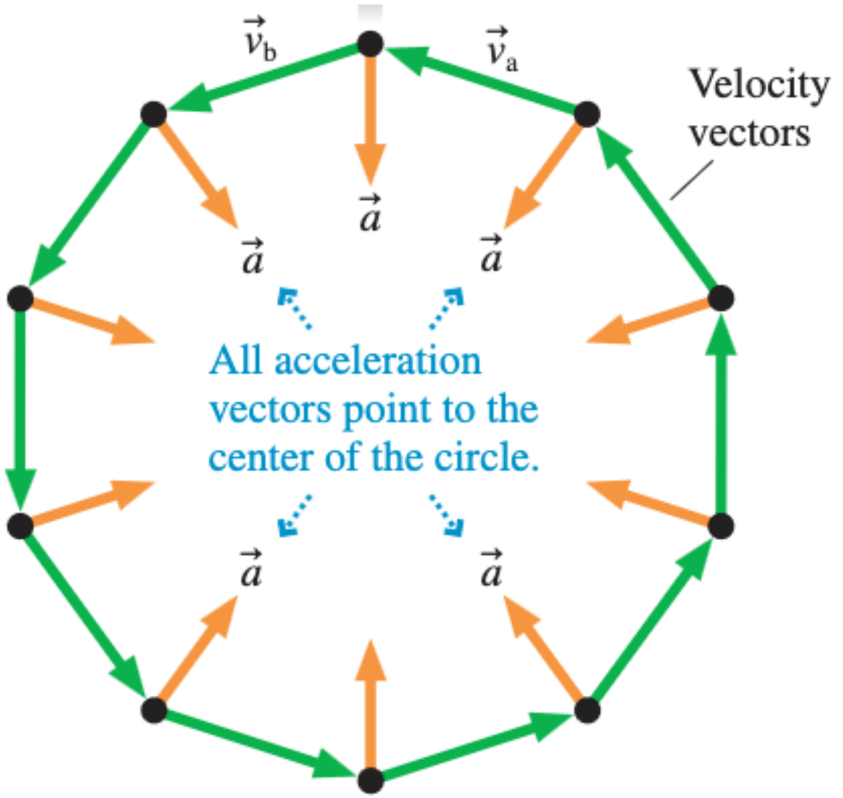
Uniform Circular Motion

- Uniform circular motion **if and only if** angular velocity ω is constant
  - $\theta_{\mathrm{f}} = \theta_{\mathrm{i}} + \omega \Delta t$
- ω related to period
  - Angular displacement one full circle: $\Delta\theta = 2\pi$ and $\Delta t = T$
  - $|\omega| = \dfrac{2\pi}{T} \longrightarrow T = \dfrac{2\pi}{|\omega|}$
  - Note: period gives the **angular speed**, need direction of motion to determine sign of ω

Uniform Circular Motion

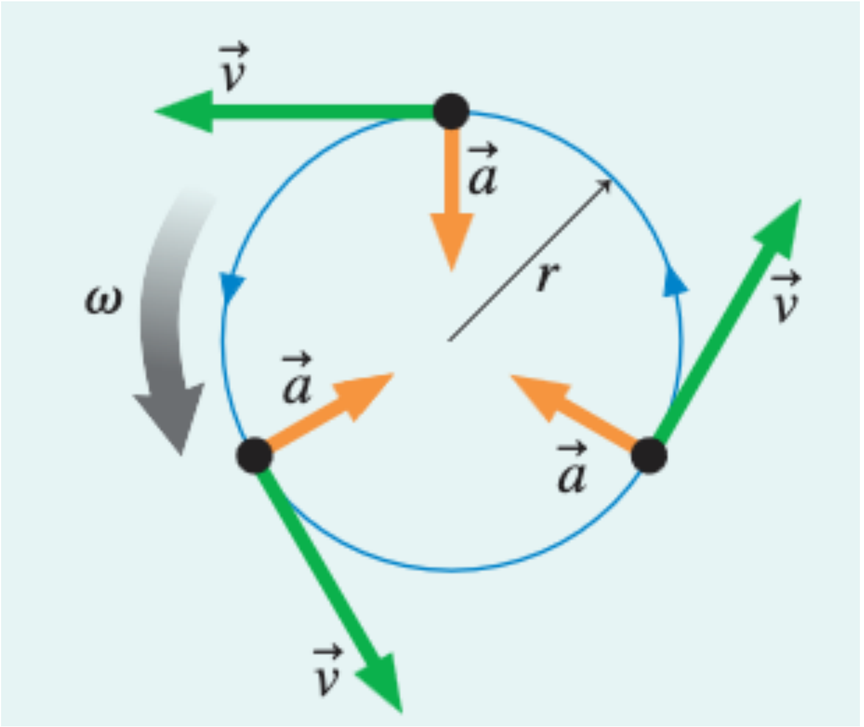
Angular velocity ω constant; so tangential velocity $v_t = r\omega$ constant

- Speed v is constant; velocity $\vec{v}$ is not
  - Velocity vector changing orientation
  - **Velocity vector changing = acceleration**
- Centripetal acceleration = acceleration when object changes direction (uniform or nonuniform motion)
  - Always points towards center of circle
  - The vector is never constant
  - The magnitude is constant for uniform circular motion



Uniform Circular Motion Model

- Motion with constant angular velocity ω
- Particle moving along a circular trajectory at constant speed
- Tangential velocity: $v_t = r\omega$
- Centripetal acceleration: $\frac{v_t^2}{r}$ or $\omega^2 r$
- Model fails if rotation isn't steady

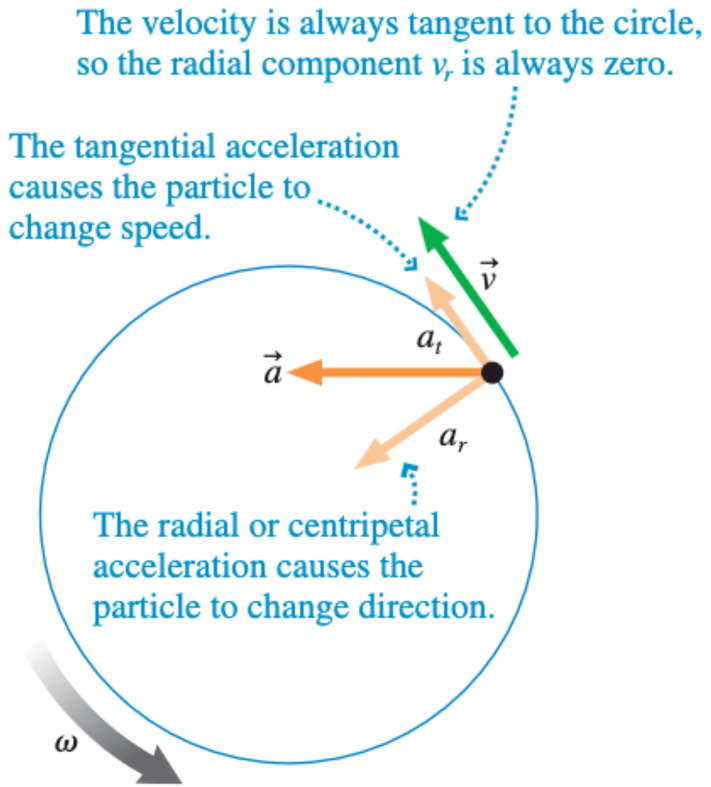


Nonuniform Circular Motion

- Circular motion with changing speed (i.e., changing angular velocity)
- Analogous to motion with linear acceleration (but with angle)
  - Angular position like linear position
  - Angular velocity like linear velocity
  - Angular acceleration like linear acceleration
- The graphical interpretation same (e.g., angular velocity is the tangent of angular position)
- Kinematic equations same form (but with angle)

Nonuniform Circular Motion

- Acceleration vector $\vec{a}$ of a does not point towards the center (points “ahead” or “behind”)
  - Radial acceleration a_r points (**radially**) towards the center of the circle, i.e., the centripetal acceleration
    - $a_r = \dfrac{v_t^2}{r} = \omega^2 r$ still valid.
  - Tangential acceleration a_t parallel to the trajectory (or equivalently $\vec{v}$).
    - Responsible for change in speed
  - Magnitude: $a = \sqrt{a_r^2 + a_t^2}$
- **For constant angular acceleration** $a_t = \dfrac{\mathrm{d}v_t}{\mathrm{d}t} = \dfrac{r\mathrm{d}\omega}{\mathrm{d}t} = r\alpha$



Angular Acceleration

- “Rate of change of angular velocity”
- $\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t}$ (lowercase Greek alpha)
- Units: $\frac{\mathrm{rad}}{\mathrm{s}^2}$
- When ω and α have:
  - same sign, speeding up
  - opposite sign, slowing down

Constant Angular Acceleration Model

- For motion with constant angular acceleration α
- Applies to particles with circular trajectories and rotating solid objects
- $\omega_{\mathrm{f}} = \omega_{\mathrm{i}} + \alpha \Delta t$
- $\theta_{\mathrm{f}} = \theta_{\mathrm{i}} + \omega_{\mathrm{i}} \Delta t + \frac{1}{2} \alpha \Delta t^2$
- $\omega_{\mathrm{f}}^2 = \omega_{\mathrm{i}}^2 + 2\alpha \Delta \theta$

Think about it

- You’re on a train moving at $22\,\mathrm{m/s}$. You walk towards the front of the train (in the direction of motion) at $1\,\mathrm{m/s}$. How fast are you going?

Reference frames

- A reference frame is a coordinate system in which position and time measurements are made from

- If you know an object's velocity in one reference frame, you can transform it into the velocity of another reference frame using a Galilean transformation.

Relative Motion

- Velocity is measured with respect to another point (“relative to”)
- Because velocity depends on what it is measured relative to another point it can change depending on the velocity of the point it is measures relative to
- Velocity is measured with respect to another point (reference frame)
- Velocity’s value changes depending on the reference point is measured from
- There is no “true” velocity (just as there is not “true” position). If measured correctly all velocities are correct.
- Velocity is actually incomplete without also knowing what reference frame it was measured with respect to (we just usually assume a stationary or “lab” frame)

Relative Motion Subscript Convention

- v_{AB} = velocity A, measured by B.
  - The first subscript is the thing moving
  - the second is who's watching
- $v_{\text{you,train}} + v_{\text{train,ground}} = v_{\text{you,ground}}$
  - The inner subscripts match and cancel; the outer ones survive
  - If you set up a problem and the inner subscripts don't match, you've combined the wrong two things
- The flip rule
  - $v_{AB} = -v_{BA}$
  - You see the train move forward at $20\,\text{m/s}$, the train sees you move backward at $20\,\text{m/s}$ (same speed opposite direction)

